Estimating Implicit Regularization in Deep Learning

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What regularization is implicit in your training procedure?

Why regularization? Explicit regularization adds penalties to the objective function that steer learning toward certain solutions—for example, those with smaller weights—in ways that can reduce overfitting and improve generalization. Training choices such as early stopping, minibatching, and dropout also bias which solutions are learned, but we often don't know exactly how [1, 2].

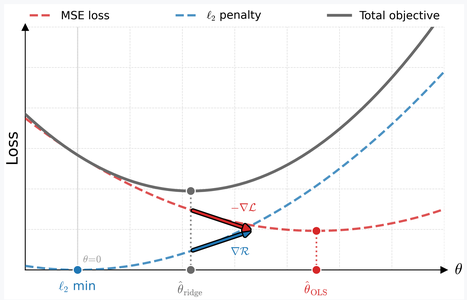
Characterizing implicit regularization. Theory characterizes some of these implicit effects, mostly in simplified settings [3, 4]. We propose an empirical framework for testing theoretical predictions and extending the study of implicit regularization to networks that are not analytically tractable.

Our approach. Work backwards: observe the learned weights and training dynamics, and estimate a parameterized regularizer that explains them. We match loss gradients or update residuals to the gradients of a proposed regularizer and solve for its parameters.

Gradient matching

Training minimizes empirical risk (ERM). We model the learned weights as a regularized ERM (RERM) solution, choosing a family $\mathcal{R}(\theta, D; \Lambda)$ and estimating its parameters Λ:

$$\hat{\theta}_{\text{ERM}} = \arg\min_{\theta} \mathcal{L}(\theta, D) \qquad \longrightarrow \qquad \hat{\theta}_{\text{RERM}} = \arg\min_{\theta} \left\{\mathcal{L}(\theta, D) + \mathcal{R}(\theta, D; \Lambda)\right\}$$



An ℓ_2 penalty shifts the solution, where the empirical loss gradient and regularizer gradient cancel.

Estimating from learned weights

At a hypothetical optimum $\hat{\theta}$ of $\mathcal{L} + \mathcal{R}$, the gradients sum to zero:

$$-\nabla_\theta \mathcal{L}(\hat{\theta}, D) = \nabla_\theta \mathcal{R}(\hat{\theta}, D; \Lambda)$$

These $p = \dim(\theta)$ equations let us solve for, or estimate, Λ:

$$\hat{\Lambda} = \arg\min_{\Lambda} \frac{1}{p} \|\nabla_\theta \mathcal{L} + \nabla_\theta \mathcal{R}_\Lambda\|_2^2$$

Example: For predictions $\hat{y} = f_\theta(X)$, MSE $\mathcal{L} = \|y - \hat{y}\|^2/n$ and $\mathcal{R} = \lambda\|\theta\|^2$ give

$$J^\top(y - \hat{y})/n = \lambda\hat{\theta}$$

where $J = \partial\hat{y}/\partial\theta$ is the prediction Jacobian, evaluated at $\hat{\theta}$.

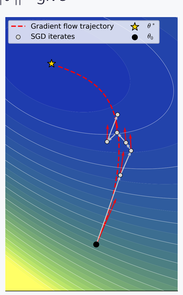
Estimating from weight updates

SGD iterates deviate from the full-batch gradient-flow path (right). For a gradient-step reference, match the update residual:

$$-\frac{\Delta\theta_t}{\eta} - \nabla_\theta \mathcal{L}(\theta_t) \;\approx\; \nabla_\theta \mathcal{R}(\theta_t; \Lambda)$$

where $\Delta\theta_t = \theta_{t+1} - \theta_t$.

Pool steps to fit a shared regularizer, or let its parameters vary over time. A zero update recovers the endpoint case.



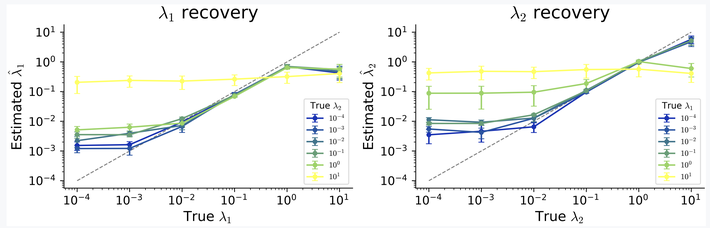
Recovering known regularization

Two ways to know that we have successfully estimated the implicit regularization:

Recover the penalty: match known regularization parameters.
Recover the weights: retrain with the estimated penalty to reproduce the learned weights.

Recovering elastic-net coefficients

Estimate both λ_1 and λ_2 in $\mathcal{R} = \lambda_1\|\theta\|_{1,\beta} + \lambda_2\|\theta\|_2^2$.



Deep ReLU networks; 6×6 grid, 10 datasets per cell. Mean ± SE; dashed = exact recovery.

Recovery is accurate at intermediate strengths. For $|\theta_j| < \beta = 10^{-3}$, smoothed ℓ_1 is quadratic, $\theta_j^2/(2\beta)$, so its gradient is proportional to the ℓ_2 gradient. Strong penalties shrink weights into this region, making λ_1 and λ_2 hard to distinguish; weak penalties give little signal.

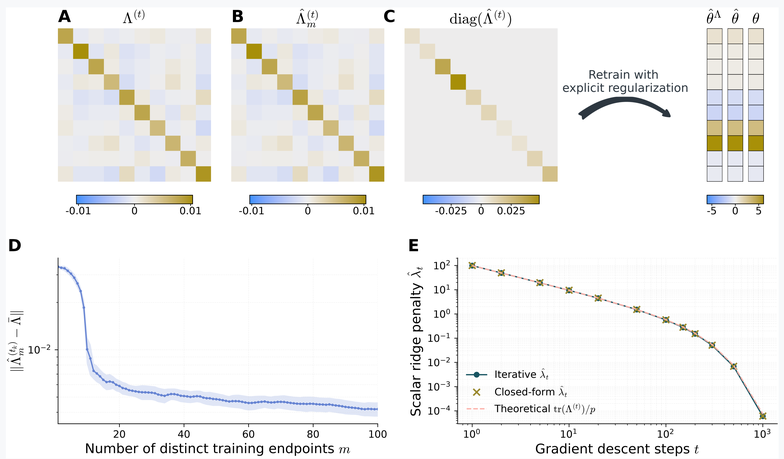
Early stopping as a quadratic penalty

The early-stopped OLS iterate minimizes $\|y - X\theta\|_2^2/n + \mathcal{R}_t(\theta)$ [5]:

$$\mathcal{R}_t(\theta) = \theta^\top \Lambda^{(t)} \theta, \qquad \Lambda^{(t)} = VS\left[(I - \eta S)^{-t} - I\right]^{-1} V^\top$$

Here $X^\top X/n = VSV^\top$, $\theta_0 = 0$, and $0 < \eta s_i < 1$.

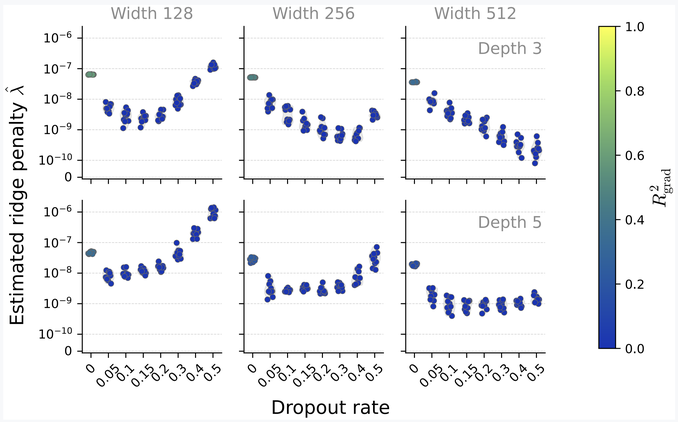
A symmetric Λ has $p(p+1)/2$ unknowns; each endpoint supplies p equations. For B, we pool diverse endpoints with shared X and t and varied responses. Diagonal fits use only p parameters.



A–B: Theory (A) and recovery from 10 endpoints at the same stopping time (B).
C: A diagonal fit from one endpoint is underparameterized, yet retraining reproduces the early-stopped weights: an equivalent regularizer, despite differing from the theoretical matrix.
D: More endpoints improve matrix estimation, even when stopping times vary.
E: The estimated scalar penalty decreases over training.

Implicit regularization in deep networks

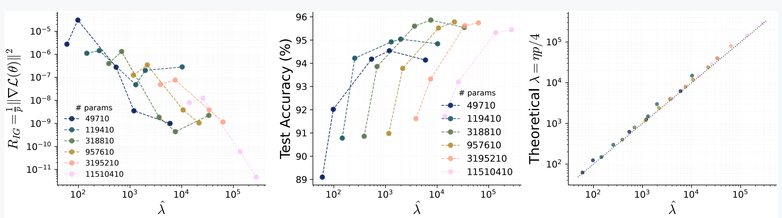
Dropout This training procedure has data-dependent regularization effects [6]. On MNIST, vary depth, width, and dropout rate; fit $\mathcal{R}(\theta, \lambda) = \lambda\|\theta\|_2^2$.



Ten seeds per setting, including zero-dropout controls. Color shows gradient fit: lighter = more explained.

Narrow networks show an upturn at high dropout; wider networks have smaller estimates. Scalar ridge summarizes part of the effect.

Implicit gradient regularization Finite GD steps approximately follow a modified gradient flow with penalty $\mathcal{R} = \frac{\lambda}{p}\|\nabla\mathcal{L}\|^2$ and strength $\lambda = \eta p/4$ [7]. This penalizes steep parts of the trajectory, biasing learning toward flatter minima associated with better generalization. We recover λ from trajectory deviations (right); stronger regularization accompanies smaller gradients and higher test accuracy (left, middle).



Future work

- **Diagnose training choices early**, before expensive full runs.
- **Study complex architectures and optimizers** through their implicit penalties.
- **Improve identification** with diverse endpoints and richer regularizer families.

References

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